Junghoon Chung

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SUMMARY

PhD candidate in Industrial and Systems Engineering specializing in machine learning, causal inference, and multimodal human behavior modeling. Experienced in developing end-to-end pipelines and evaluation frameworks that translate complex data into actionable, user-centered insights. Skilled in applying AI/ML methods, statistical analysis, and experimental design to real-world problems, with publications in human factors and ergonomics. Seeking a research or data science internship to develop adaptive, data-driven solutions.

EDUCATION

University of Washington, Seattle Sep 2025 – Jun 2029 (Expected)
Ph.D in Industrial and Systems Engineering, *Dean's Fellowship Recipient*

Virginia Tech, Blacksburg Aug 2023 – May 2025
M.S in Industrial and Systems Engineering (Thesis Track), *GPA 3.89/4.0*

Yonsei University, Seoul Mar 2017 – Feb 2023
B.S in Industrial Engineering, *GPA 3.52/4.0*, *Student Council President*

EXPERIENCE

Graduate Research Assistant, Human Automation Systems Lab, University of Washington Sep 2025 – Present

- Built ML models to assess vigilance using multimodal physiological time-series data (ECG, eye tracking, BVP, EDA, HR), and developed preprocessing pipelines for heterogeneous sensor streams using Python and scikit-learn
- Engineered 30+ physiological and contextual features across 10,000+ observations per participant, improving predictive performance by ~20% through feature engineering and model refinement

Graduate Research Assistant, Occupational Ergonomics and Biomechanics Lab, Virginia Tech Aug 2023 – May 2025

- Developed end-to-end data pipeline to collect, preprocess, and analyze multimodal datasets (EMG, EEG, IMU, motion capture, behavioral logs) across longitudinal studies (3 weeks/participant, 100,000+ records)
- Utilized a zero-shot NLP classifier for unstructured work-context text, built time-series forecasting models incorporating contextual variables, and improved prediction accuracy through data segmentation and clustering

KEY PROJECTS

Human Vigilance Modeling with Causal Inference | *University of Washington*
Built and compared predictive models of alertness using multimodal physiological data (ECG, BVP, EDA, HR, eye tracking), evaluating how time-series aggregation windows (5, 10, 30, 60 seconds) affect model performance. Examined how multimodal signals relate to sustained attention via task reaction time, across 10,000+ observations per participant.

Driver Response Modeling for V2X Pedestrian Alerts | *City of Bellevue / University of Washington*
Designing a field study to collect physiological sensor data (E4 Empatica: EDA, BVP, HR), NASA-TLX workload ratings, and vehicle kinematics to model driver responses to V2X crosswalk pedestrian alerts. Data collection planned for October.

Multi-Stage Behavioral Framework for Driver Takeover | *University of Washington*
Proposing a framework decomposing automated-driving takeover into three interpretable stages (Attention Reorientation, Control Acquisition, Control Stabilization) using eye-tracking and simulator data, designed to diagnose where drivers differ rather than only predict takeover success.

NudgeWatch, Adaptive Sit-Stand Nudging System | *University of Washington*
Built a full-stack Apple Watch intervention system (Swift, Python/FastAPI) with real-time posture detection and an active-learning nudge engine. Ran a 5-participant pilot combining sensor logs with experience-sampling feedback, and found the adaptive loop measurably improved outcomes, though adaptation was rarely correctly perceived by users. Portfolio: Junghoon-Chung.github.io/#nudgewatch

Context-Aware Sit-Stand Intervention | *Virginia Tech*
Segmented 3+ weeks of longitudinal behavioral data (100,000+ records/participant) into personalized usage clusters, then fit per-cluster time-series models to forecast posture transitions. The personalized approach outperformed a single population-level model across all participant groups.

SKILLS

Programming: Python, SQL (MySQL), Swift (WatchOS)
ML/DL: Scikit-learn, TensorFlow, PyTorch
Data Analysis: Statistical modeling, Causal inference, Experimental design, Time-series analysis
Communication: Technical reporting, Cross-functional collaboration

AWARDS

Dean's Fellowship, College of Engineering — University of Washington, 2025–2029
Social-Innovation Capability Video Contest 2nd Place, Data mining — Yonsei University, 2022
Merit-based Scholarship (Academic Excellence) — Yonsei University, 2021